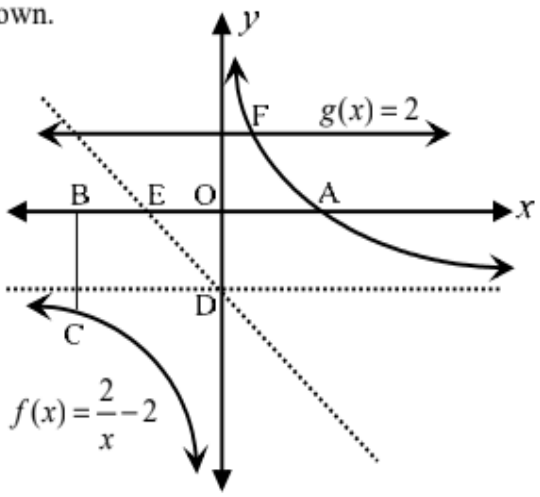
Worksheet on Graph Application

1. The graph of $f(x)=\frac{2}{x}-2$ and $g(x)=2$ is shown.



Determine:

(1) the length of OA.

(2) the length of BC if OB = 4 units

(3) the length of OD and OE

(4) the coordinates of F

(5) the values of x for which $f(x)\geq 0$

(6) the values of x for which $f(x)<0$

(7) the values of x for which $f(x)>-2$

(8) the values of x for which $f(x)>g(x)$

2. The graph of $f(x)=\left(\frac{1}{2}\right)^x-2$ is shown.

Determine:

(1) the length of OA and CD.

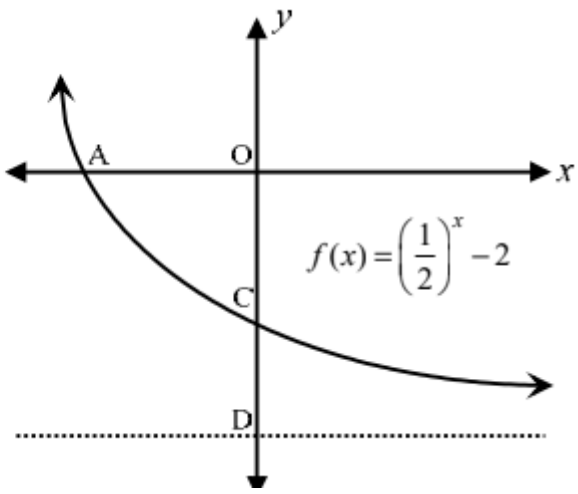
(2) the values of x for which $f(x)>0$

(3) the values of x for which $f(x)\le 0$

(4) the values of x for which $f(x)>-2$

(5) the values of x for which $f(x)>-1$

(6) the values of x for which $f(x)\le -1$



3.

The graph of $f(x) = a.b^x + q$ and $g(x) = \frac{a}{x} + 6$ cut the x-axis at A. The graph of f passes through the point $(-2\,;-12)$ and cuts the y-axis at 4.

Determine:

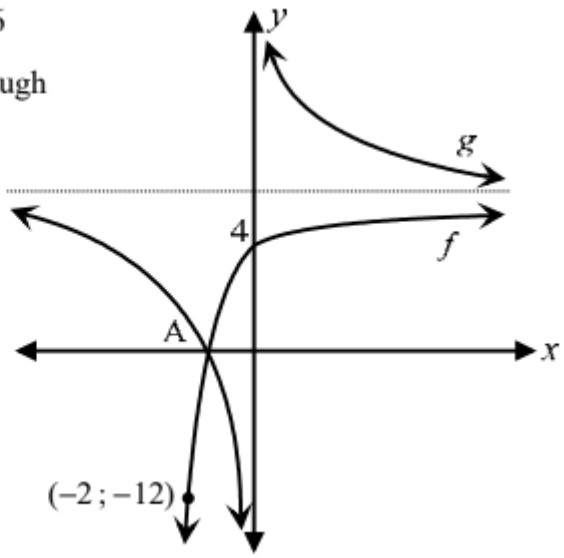
(1) the range of f

(2) the equation of f

(3) the equation of g

(4) the equation of h if the graph of f is reflected in the x-axis

(5) the values of x for which f is increasing



4. In the diagram below, the graphs of f and g are shown.
The graphs intersect at $(1;3)$. Determine:

(1) the equation of f

(2) the coordinates of the turning point of f.

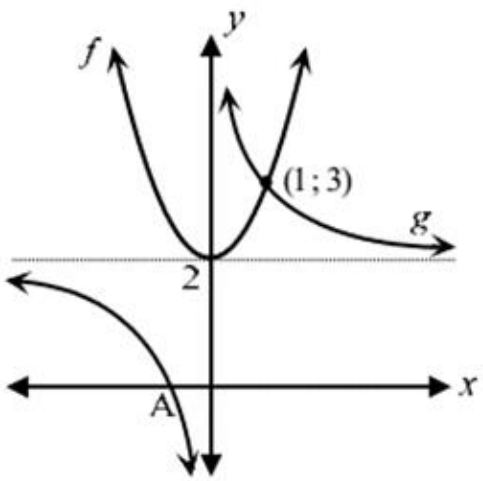
(3) the equation of the axis of symmetry of f.

(4) the equation of the horizontal asymptote of g

(5) the equation of g and the coordinates of A

(6) the domain and range of f and g

(7) the values of x for which f is decreasing?



5. The graph of $f(x) = mx + c$ and $g(x) = ax^2 + b$ intersect at D and on the x-axis at A.
H and T lie on line f. Determine:

(1) the equation of f

(2) the coordinates of A and hence C

(3) the equation of g

(4) the length of AC and OB

(5) the length of DE and DF

(6) the length of ST

(7) the length of OM if $QR = \frac{45}{4}$

(8) the values of x for which $g(x) \geq 0$

(9) the values of x for which $g(x) < f(x)$

